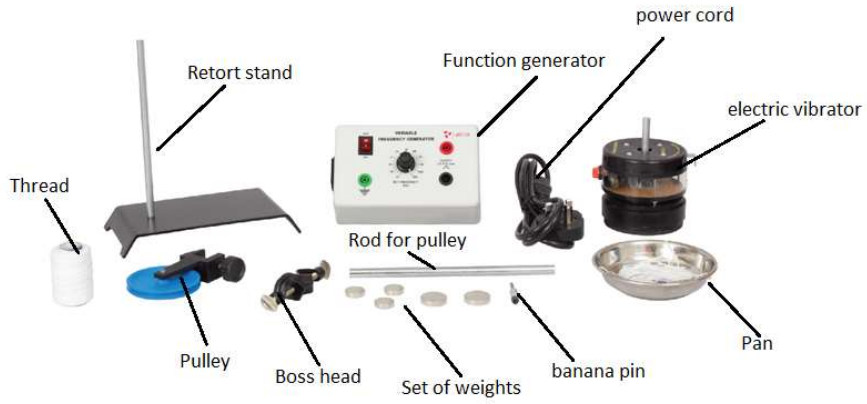
Melde's Experiment

Aim:

a. To obtain the relationship between frequency, tension and length of a string.
b. To determine the mass per unit length of a string.

Apparatus:

Rod for pulley, Pulley, Set of Weights, Pan, Frequency/Signal Generator, Retort Stand, Power Cord, Boss head, Meter scale, Weighing scale, Thread or String.



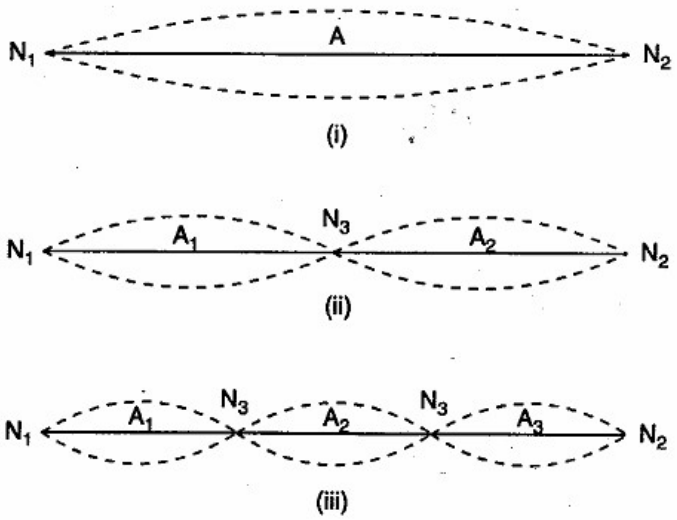
Theory:

The arrangement in which the steel rod executes vibrations in a direction **perpendicular** to the **direction** of propagation of the wave (or the length of the string) is called **transverse arrangement**. The arrangement in which the steel rod executes vibrations in a direction **parallel** to the **direction** of propagation of the wave (or the length of the string) is called **longitudinal arrangement**.

When an alternating current flows through the solenoid coil, the steel rod is magnetized with its magnetic polarity being reversed after each half cycle of alternating currents. Due to the interaction between the magnetism of the rod and the horse-shoe magnet, the steel rod R starts vibrating with frequency from the signal generator. The rod executes forced vibrations but on adjusting the length of the rod, which is free to vibrate, resonance can be obtained when the normal vibration frequency of the rod becomes exactly the same as the frequency from the signal generator. Consequently, the vigorous vibrations occur at the free end of the rod and the vibrator maintains continuous vibrations at the expense of energy supplied from the signal generator.

When a string stretched between two fixed points is plucked in a direction at right angles to its length, stationary transverse vibrations are set up in it. The string may vibrate as a whole in one segment as shown in fig (i). In this case, the frequency of the sound produced by the string is the lowest and is called fundamental. It may vibrate in two or more segments as shown in fig (ii) and (iii). In these cases, the

frequency of the sound produced will be double and three times the fundamental frequency respectively. These frequencies are known as overtones and harmonics.



The fixed points like N_1, N_2, N_3, … etc., where the amplitude is zero but the strain is maximum are called nodes and points like A_1, A_2, A_3, …etc., where the amplitude is maximum but the strain is the least are called antinodes.

When the alternating current is passed, the steel rod executes vibrations which pass over to the thread. The wave travelling along the thread is reflected by the pulley. Thus, the stationary waves are formed in the thread with the pulley and the end of the vibrator as nodes. Depending on the length of the thread and the tension, i.e., the mass suspended in the pan, the thread may vibrate in several loops. The length of the thread between the vibrator rod and the pulley is adjusted until the thread vibrates in a definite number of loops with maximum amplitude at the antinodes i.e., resonance takes place and sharp nodes and antinodes are formed. In this arrangement, the frequency of the thread is the same as that of the electric vibrator which is through a frequency or signal generator. Therefore,

Frequency of the electric vibrator = Frequency of the thread = f

Longitudinal Arrangement: The time, during which the steel rod completes one vibration, the thread completes half a vibration. Hence the frequency of the thread is half that of the steel rod. If the same length l of the thread vibrates in one loop under a tension T, then the frequency of the vibration of the thread is given by the relation

$$n = 2 \times \text{Frequency of the thread}$$

The frequency in the longitudinal arrangement is given by the relation

$$f = \frac{1}{l}\sqrt{\frac{T}{\mu}}$$

where l is the length of the thread in the fundamental mode of vibration

T = Mg is the tension produced in the thread.

μ is the mass per unit length of the thread.

Transverse Arrangement: The time during which the steel rod completes one vibration, the thread also completes one vibration. If l is the length between two consecutive nodes, T the tension present in the thread and m its mass per unit length, then the frequency from the signal generator in the transverse arrangement is given by the relation

$$f = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$$

where l is the length of the thread in the fundamental mode of vibration
T = Mg is the tension produced in the thread.
μ is the mass per unit length of the thread.

We will focus only on the Transverse arrangement in this manual.

There are two experiments that need to be done to verify the relationship between frequency, length and tension in the string. We will vary the frequency with mass keeping length fixed and vary the frequency with length keeping the mass fixed.

Procedure:

1. Connect the frequency/signal generator with the mains, then connect the electric vibrator with the generator using patch cords.
2. Mount the rod on the retort stand using the boss head.
3. Clamp the pulley on the rod at a height approximately equal to the electric vibrator.
4. Insert the banana pin into the electric vibrator, bind one end of the thread with the electric vibrator and the other end to the light pan.
5. Suspend the pan freely after passing over a frictionless pulley mounted on the retort stand. The height of the pulley is such that the cord remains horizontal.
6. Switch on the frequency generator, and set the frequency so that the sharp nodes and antinodes produce in the thread for the fundamental mode.
7. Note down the readings in Table 1 which will record frequencies that result in a fundamental mode for different masses keeping the length of the thread fixed.
8. Plot f^2 versus T. A straight line verifies that frequency varies directly with mass.
9. Note down the readings in Table 2 which will record frequencies that result in a fundamental mode for different lengths keeping the mass fixed.
10. Plot f^2 versus $1/l^2$. A straight line verifies that frequency varies inversely with the length of the string.
11. Find the mass per unit length (μ) from the slope of the graph of f^2 versus $1/l^2$.

Observations:

Mass of the pan (M_p) = ---------- gm

Table 1 (Fix the length of the string and vary the mass)

S. No.	Length of thread l (cm)	Mass placed in the pan m (gm)	Tension $T = (m + M_p)g$	Frequency (f)	f^2
1		0			
2		10			
3					
4					
5					

Plot f^2 versus T.

Table 2 (Fix the mass and vary the length)

S. No.	Length of thread l (cm)	Mass placed in the pan m (gm)	Tension $T = (m + M_p)g$	Frequency (f)	f^2
1	150	10			
2	130	10			
3					
4					
5					

Plot f^2 versus $1/l^2$.

Result:

(i) The frequency varies _____ with the Tension in the string.
(ii) The frequency varies _____ with the length of the string.
(iii) The mass per unit length is _____ g/cm.

Precautions and sources of error:

1. Flexible, inextensible thread of uniform thickness should be used.
2. The pulley should be frictionless and the thread between the vibrator and the pulley should be perfectly horizontal.
3. Sharply defined nodes and antinodes should be noted. The loops formed in the thread with maximum amplitude at the antinodes should be noted.
4. Do not put too much load in the pan.